Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)
The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

$$\texttt{daisy}, \quad \texttt{dandelion}, \quad \texttt{roses}, \quad \texttt{sunflowers}, \quad \texttt{tulips}.$$

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

Observations: Dataset Description

`tf_flowers` contains 3,670 RGB images of five classes. All images were resized to $224 \times 224 \times 3$ and cached once. A fixed, stratified 70/15/15 split (seed 42) gives **2,569 training, 550 validation and 551 test images**, reused for all 17 experiments. The test split has 135 dandelion, 95 daisy, 120 tulips, 105 sunflowers and 96 roses images (class order of the dataset: dandelion, daisy, tulips, sunflowers, roses). Images vary strongly in scale, viewpoint, illumination, background clutter and flower size (Fig. 1).

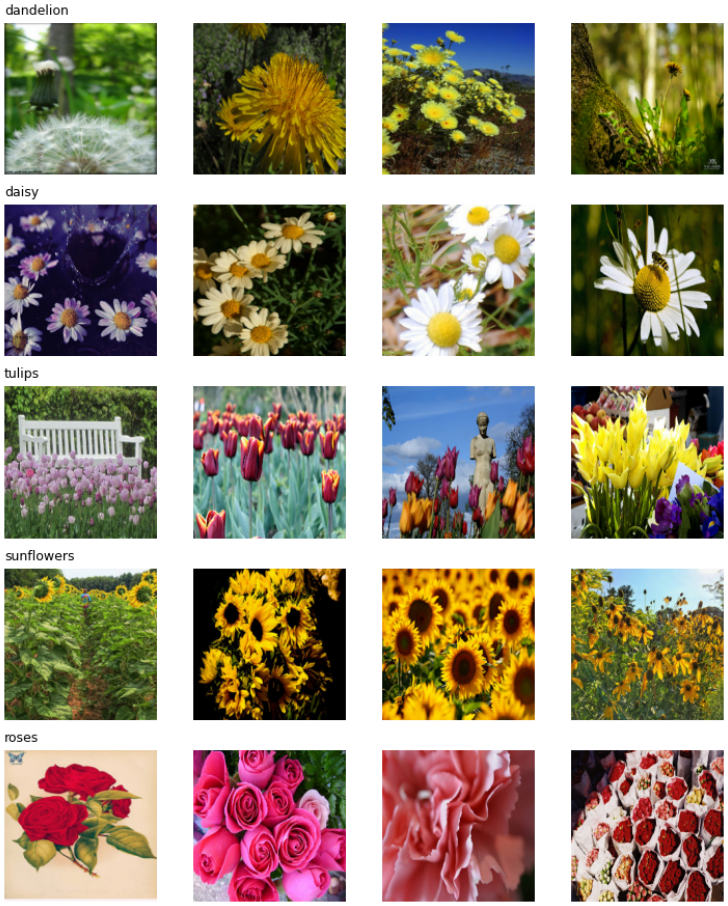


Figure 1: Sample training images of the five classes (4 per class).

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

Observations: ResNet50 Architecture

Table 1: Measured ResNet50 stage hierarchy (ImageNet weights, $224 \times 224 \times 3$ input). All four positions match the table of Section 7.

Stage	Position	Resolution	Channels	Residual blocks	Parameters	Level of features
Stem + MaxPool	–	56×56	64	0	9,728	early: edges, colours, orientations
conv2_x	P_1	56×56	256	3	220,032	early: boundaries, textures
conv3_x	P_2	28×28	512	4	1,230,336	early/intermediate: local patterns
conv4_x	P_3	14×14	1024	6	7,129,088	intermediate: flower parts, shapes
conv5_x	P_4	7×7	2048	3	14,998,528	late: class-discriminative semantics

As depth increases the resolution falls from 56×56 to 7×7 while the channels rise from 256 to 2048. ResNet50 contains **16 residual (Add) connections** (3+4+6+3 bottleneck blocks); conv5_x alone holds 14,998,528 of the 23,587,712 backbone parameters (about 64%). The baseline (pretrained ResNet50 $\rightarrow$ global average pooling $\rightarrow$ Dropout 0.3 $\rightarrow$ Dense(5)) has **23,597,957** parameters, of which 10,245 belong to the new classification head.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1 : \text{ after conv2_x},$$

$$P_2 : \text{ after conv3_x},$$

$$P_3 : \text{ after conv4_x},$$

$$P_4 : \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

Which feature channels are important?

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\text{max}} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\text{max}}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

Which spatial locations are important?

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

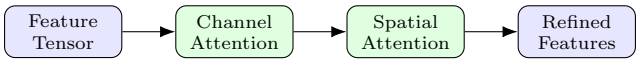
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

Observations: Attention Mechanisms and Proposed Experimental Architecture

Each module takes $F \in \mathbb{R}^{H \times W \times C}$ and returns a tensor of the same shape; it is inserted between two ResNet stages ("previous stage → attention → next stage").

- **Channel attention (= SE):** GAP → Dense($C/16$, ReLU) → Dense(C, sigmoid), gating F channel-wise. Because the PDF formulation is identical to SE, it is reported as "Channel / SE".
- **Spatial attention:** channel-wise mean and max maps → 7×7 convolution → sigmoid (99 parameters).
- **Channel→Spatial:** the two above applied sequentially; **Spatial→Channel** reverses the order.
- **Self-attention:** $Q = XW_Q$, $K = XW_K$ with $d = C/8$, $V = XW_V$, $A = \text{softmax}(QK^\top/\sqrt{d})$, output $F + \gamma\,\text{reshape}(AV)$ with γ initialised to 0.
- **Combined (A5):** Channel→Spatial→Self-attention.
- **ECA:** 1-D convolution over the channel descriptor with adaptive kernel size ($k = 5$ at $C = 1024$), 5 parameters.
- **CBAM:** channel gate from average- and max-pooled descriptors through a shared MLP, followed by a 7×7 spatial gate.
- **Coordinate attention:** direction-aware pooling along H and W with a shared 1×1 transform.

Table 2: Observation 1 – feature dimensions immediately before and after the attention block (none of the modules changes H, W or C).

Models	Position	Input to attention	Output of attention
A1/A2/A3/A4/A5, D1, E1, E2, F1, F2, F3 (single P_3 module)	P_3	$14 \times 14 \times 1024$	$14 \times 14 \times 1024$
B1	P_1	$56 \times 56 \times 256$	$56 \times 56 \times 256$
B2 (and C2)	P_2	$28 \times 28 \times 512$	$28 \times 28 \times 512$
B4 (and C1, C2)	P_4	$7 \times 7 \times 2048$	$7 \times 7 \times 2048$

Table 3: Registry of the 17 trained models.

ID	Attention	Position(s)	Notes
A0	none	–	pretrained ResNet50 → GAP → Dropout → Dense(5)
A1	Channel (= SE)	P_3	GAP, FC(C/16), ReLU, FC(C), sigmoid
A2	Spatial	P_3	mean/max over channels, 7×7 conv, sigmoid
A3 (= B3)	Channel→Spatial	P_3	selected model
A4	Self-attention	P_3	Q, K with $d = C/8$, V with C, learnable residual γ (init. 0)
A5	Channel→Spatial→Self-attn	P_3	combined
B1, B2, B4	Channel→Spatial	P_1, P_2, P_4	position ablation
C1, C2	Channel→Spatial	P_3+P_4; P_2+P_3+P_4	multi-stage
D1	Spatial→Channel	P_3	order ablation
E1, E2	Channel→Spatial	P_3	spatial kernel $k = 3$, $k = 5$ ($k = 7$ is A3)
F1	ECA	P_3	adaptive 1-D kernel ($k = 5$ at $C = 1024$), 5 parameters
F2	CBAM	P_3	avg+max shared MLP channel gate, then 7×7 spatial gate
F3	Coordinate Attention	P_3	direction-aware pooling along H and W

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

Observations: Attention-Type Ablation (Experiment 1)

Table 4: Experiment 1 – attention-type ablation at P_3 (test split, 551 images).

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
A0 Baseline	0.9002	0.9116	0.9026	0.9033	23,597,957	0	8.08	7.75	3.6
A1 Channel (SE)	0.9129	0.9159	0.9105	0.9123	23,730,117	132,160	8.06	7.75	3.7
A2 Spatial	0.9419	0.9453	0.9396	0.9419	23,598,056	99	8.17	7.75	3.7
A3 Channel+Spatial	0.9437	0.9469	0.9389	0.9420	23,730,216	132,259	8.37	7.75	3.8
A4 Self-Attention	0.9419	0.9432	0.9400	0.9412	24,908,678	1,310,721	7.62	8.35	4.1
A5 Combined (C+S+SA)	0.9274	0.9347	0.9237	0.9266	25,040,937	1,442,980	7.93	8.36	4.4

Relative to the baseline (accuracy 0.9002, macro F1 0.9033):

- **Spatial (A2), Channel→Spatial (A3) and Self-attention (A4)** give the clearest gains: +4.2, +4.4 and +4.2 accuracy points (+3.9, +3.9, +3.8 macro-F1 points), each beyond the ±2.5-point interval. A2, A3 and A4 are practically tied with each other.
- **Channel/SE (A1)** gives only +1.3 points, which is within noise.
- **Combined (A5)** reaches 0.9274 (+2.7 points), *lower* than A3 and A4 although it contains both: more attention in one place is not better here.

The A2 result is notable: a spatial module with **99 parameters** matches self-attention with 1,310,721 parameters. Sections 28 and 29 show why this gain should be interpreted cautiously.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

Channel Attention → Spatial Attention

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

How does the location of attention influence performance?

Observations: Attention-Position Ablation (Experiment 2)

Table 5: Experiment 2 – Channel→Spatial attention at P_1 (B1), P_2 (B2), P_3 (B3 = A3) and P_4 (B4).

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
B1 C+S @ P_1	0.9129	0.9201	0.9110	0.9119	23,606,520	8,563	8.16	7.76	5.3
B2 C+S @ P_2	0.9328	0.9349	0.9315	0.9326	23,631,368	33,411	8.03	7.75	4.5
A3 Channel+Spatial	0.9437	0.9469	0.9389	0.9420	23,730,216	132,259	8.37	7.75	3.8
B4 C+S @ P_4	0.9365	0.9361	0.9378	0.9351	24,124,520	526,563	7.87	7.75	3.5

P_3 is best (0.9437 / 0.9420), followed by P_4 (0.9365), P_2 (0.9328) and P_1 (0.9129, only +1.3 points over the baseline). The P_2–P_4 differences are about one standard error, so only two statements are supported: attention at the high-resolution P_1 is clearly less useful than at P_3 (−3.1 points) and also costs the most (5.3 min estimated training time versus 3.8 min at P_3), and intermediate/late positions work at least as well as each other. Attention parameters

grow with channel width (8,563 at P_1, 33,411 at P_2, 132,259 at P_3, 526,563 at P_4) since the channel MLP scales with C^2.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

Observations: Multiple Attention Positions (Experiment 3)

Table 6: Experiment 3 – single versus multiple attention positions (Channel→Spatial).

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
A3 Channel+Spatial	0.9437	0.9469	0.9389	0.9420	23,730,216	132,259	8.37	7.75	3.8
B4 C+S @ P_4	0.9365	0.9361	0.9378	0.9351	24,124,520	526,563	7.87	7.75	3.5
C1 C+S @ P_3+P_4	0.9292	0.9279	0.9306	0.9286	24,256,779	658,822	7.81	7.75	3.9
C2 C+S @ P_2+P_3+P_4	0.9074	0.9147	0.9108	0.9079	24,290,190	692,233	8.41	7.76	4.7

Applying attention at more stages **did not help**: P_3+P_4 is 1.5 accuracy points below P_3 alone, and P_2+P_3+P_4 is 3.6 points below, while adding 526,563 and 559,974 attention parameters, and increasing estimated training time from 3.8 to 3.9 and 4.7 min. The three-stage model also shows a large train–validation gap (0.0825 versus 0.0412 for A3). The answer to "Is attention at more locations necessarily more effective?" is therefore *no*.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \to M_c \to M_s$$

should be compared with

$$F \to M_s \to M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

Observations: Attention Order (Experiment 4)

Table 7: Experiment 4 – attention order at P_3.

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
A3 Channel+Spatial	0.9437	0.9469	0.9389	0.9420	23,730,216	132,259	8.37	7.75	3.8
D1 Spatial→Channel	0.9328	0.9306	0.9338	0.9319	23,730,216	132,259	7.97	7.75	3.8

Channel→Spatial is 1.1 points better than Spatial→Channel (0.9437 vs 0.9328), which is below two standard errors: **no reliable order effect**. The spatial maps of the two orders are very similar (per-image correlation 0.93–0.99, Fig. 2).

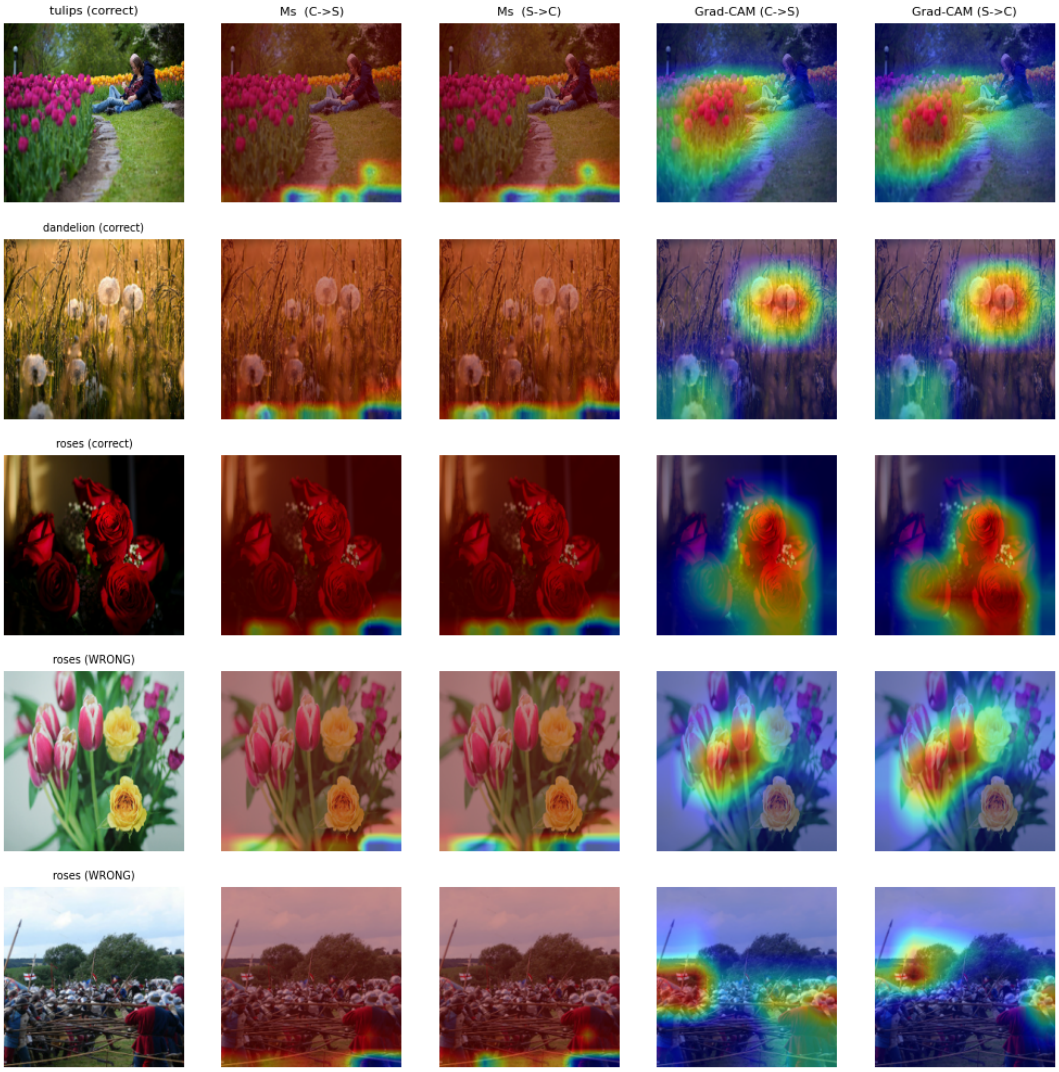


Figure 2: Experiment 4: spatial attention maps and Grad-CAM for Channel→Spatial (C→S) and Spatial→Channel (S→C) at P_3.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	0.9056	0.9056	8.01 ms
5×5	0.9310	0.9302	8.17 ms
7×7	0.9437	0.9420	8.37 ms

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

Observations: Spatial Kernel Size (Experiment 5)

Table 8: Experiment 5 – spatial kernel size k (Channel→Spatial at P_3; k=7 is A3).

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
E1 C+S, k=3	0.9056	0.9044	0.9109	0.9056	23,730,136	132,179	8.01	7.75	3.8
E2 C+S, k=5	0.9310	0.9319	0.9303	0.9302	23,730,168	132,211	8.17	7.75	3.8
A3 Channel+Spatial	0.9437	0.9469	0.9389	0.9420	23,730,216	132,259	8.37	7.75	3.8

Accuracy rises monotonically with the kernel: 3×3: 0.9056, 5×5: 0.9310, 7×7: 0.9437. The 3×3 model is essentially the baseline, while 7×7 is 3.8 points better (beyond the noise interval). However, the learned maps do not show a richer use of context for larger kernels (Fig. 3): the 3×3 maps are the most varied (mean 0.787, std 0.224, suppressing the left/bottom border and some textured regions), whereas the 5×5 maps are almost saturated (mean 0.972, std 0.059) and the 7×7 maps (mean 0.966, std 0.131) are ≈ 1 everywhere except a band along the bottom border. So a larger kernel is associated with higher accuracy, but the maps do not demonstrate that the gain comes from extra spatial context.

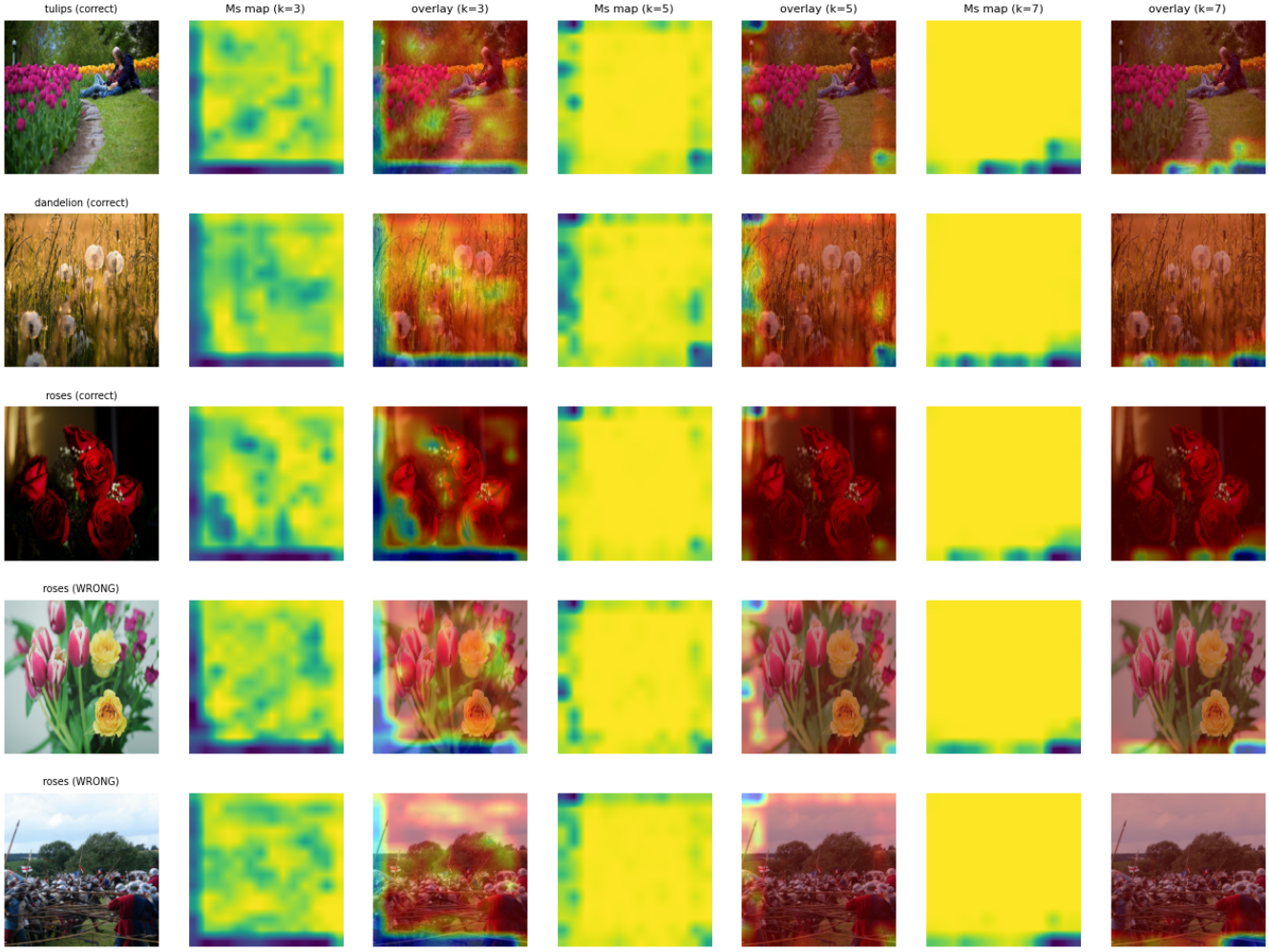


Figure 3: Experiment 5: spatial attention maps and overlays for $k = 3, 5, 7$ at P_3.

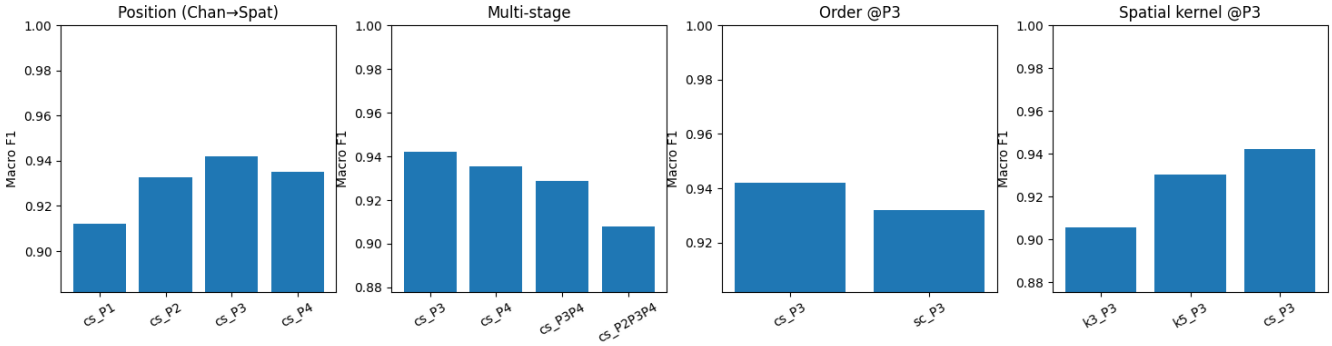


Figure 4: Macro F1 of the position, multi-stage, order and kernel-size ablations.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);

- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy	Precision	Recall	Macro F1	Parameters	Inference Time
No Attention	0.9002	0.9116	0.9026	0.9033	23,597,957	8.08 ms
Channel / SE	0.9129	0.9159	0.9105	0.9123	23,730,117	8.06 ms
Spatial Attention	0.9419	0.9453	0.9396	0.9419	23,598,056	8.17 ms
ECA	0.9238	0.9217	0.9246	0.9222	23,597,962	7.77 ms
CBAM	0.9201	0.9206	0.9203	0.9194	23,730,216	7.59 ms
Self-Attention	0.9419	0.9432	0.9400	0.9412	24,908,678	7.62 ms
Additional Mechanism (Coordinate Attention)	0.9383	0.9379	0.9373	0.9375	23,698,469	8.32 ms

Observations: Performance Results

The additional-mechanism table of Section 19.1 is filled in above; the main result table of Section 23 is filled in below. The full comparison, including complexity measures, is given here.

Table 9: Additional-mechanism comparison at P_3 (Channel/SE, Spatial, ECA, CBAM, Self-Attention, Coordinate Attention).

Model	Acc.	Prec.	Recall	Macro F1	Total params	Attn params	Inf. ms/img	GFLOPs	Train min
A0 Baseline	0.9002	0.9116	0.9026	0.9033	23,597,957	0	8.08	7.75	3.6
A1 Channel (SE)	0.9129	0.9159	0.9105	0.9123	23,730,117	132,160	8.06	7.75	3.7
A2 Spatial	0.9419	0.9453	0.9396	0.9419	23,598,056	99	8.17	7.75	3.7
F1 ECA	0.9238	0.9217	0.9246	0.9222	23,597,962	5	7.77	7.75	3.8
F2 CBAM	0.9201	0.9206	0.9203	0.9194	23,730,216	132,259	7.59	7.75	3.8
A4 Self-Attention	0.9419	0.9432	0.9400	0.9412	24,908,678	1,310,721	7.62	8.35	4.1
F3 Coordinate Att.	0.9383	0.9379	0.9373	0.9375	23,698,469	100,512	8.32	7.76	3.8

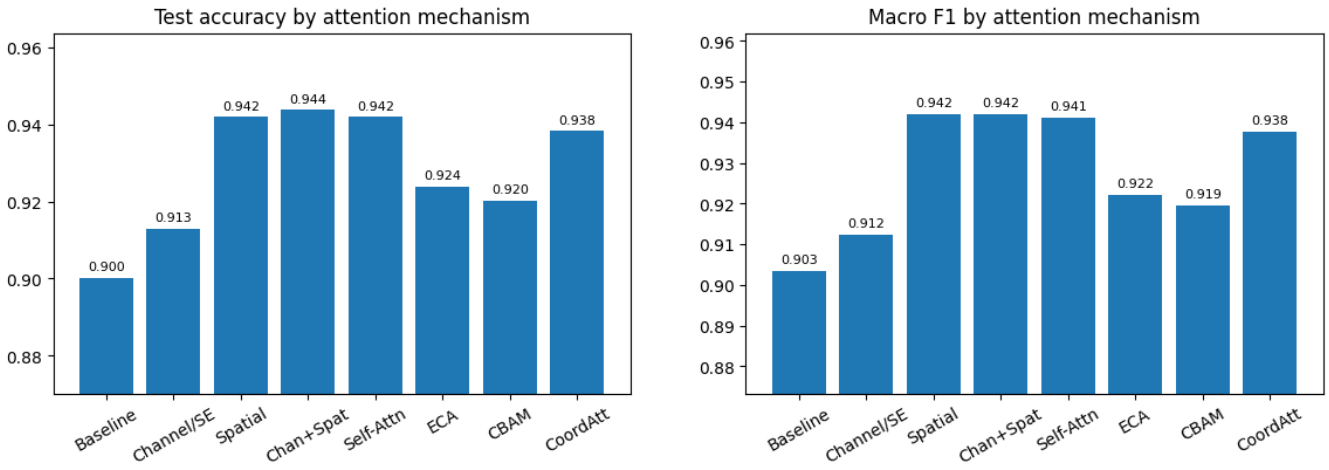


Figure 5: Test accuracy and macro F1 by attention mechanism (all at P_3).

Ranking by accuracy at P_3: Channel→Spatial 0.9437 > Spatial = Self-attention 0.9419 > Coordinate 0.9383 > ECA 0.9238 > CBAM 0.9201 > Channel/SE 0.9129 > baseline 0.9002. Macro F1 follows the same ordering (0.942, 0.942,

0.941, 0.938, 0.922, 0.919, 0.912, 0.903). Precision and recall are balanced for every model (differences of at most 0.011), so there is no sign of a model trading one for the other.

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

Observations: Additional Attention-Mechanism Comparison

Three additional mechanisms were compared at P_3: ECA, CBAM and coordinate attention (SE is the Channel module). Coordinate attention (100,512 parameters, 0.9383) is the strongest of them and close to the best models; ECA (5 parameters) improves the baseline by 2.4 points; CBAM (132,259 parameters, 0.9201) gives +2.0 points and is *worse* than the spatial module alone, which it contains. Answers to the questions of Section 19.2:

1. *Accuracy:* from 0.9002 (baseline) to 0.9437 (Channel→Spatial).
2. *Macro F1:* from 0.9033 to 0.9420, same ordering.
3. *Parameters:* 5 (ECA), 99 (spatial), 100,512 (coordinate), 132,160–132,259 (SE, Channel→Spatial, CBAM), 1,310,721 (self-attention), 1,442,980 (combined).
4. *Inference time:* 7.59–8.37 ms/image with no systematic relation to cost (noise); GFLOPs rise only for self-attention (+0.60).
5. *Convergence:* not accelerated; the train–validation gap and late validation-loss rise are smaller than the baseline's.
6. *Localisation:* only marginal changes (coverage, spread, centre mass differ by about 0.02–0.1).
7. *Complexity vs performance:* more complexity does not always help (combined < Channel→Spatial; P_2+P_3+P_4 < P_3; CBAM < spatial).
8. *Lightweight vs heavy:* yes: 99-parameter spatial attention (0.9419) ≈ 1.31 M-parameter self-attention (0.9419), and 5-parameter ECA achieves about half of the best gain.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

Observations: Experimental Setup

All models share the same protocol (Section 34): same split, ImageNet preprocessing (`resnet50.preprocess_input`), augmentation (random flip, rotation ±5% of a turn, zoom 10%, translation 10%, contrast 10%), seed 42, Adam optimiser, batch size 32, categorical cross-entropy and the same two-phase transfer-learning procedure: **Phase 1** (5 epochs, learning rate 10^{-3}): backbone frozen, only attention and head trained; **Phase 2** (10 epochs, learning rate 10^{-4}): conv5_x unfrozen (BatchNorm layers kept frozen). No early stopping or checkpoint selection was used, so all results are *last-epoch* (epoch 15) results. Inference time is the median over three passes on 256 test images (batch 32) on the same GPU; GFLOPs come from the TensorFlow profiler; training time is estimated from the measured median step time scaled to the full protocol.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	Position	Accuracy	Precision	Recall	F1	Parameters	Inference Time
Baseline	–	0.9002	0.9116	0.9026	0.9033	23,597,957	8.08 ms
Channel Attention	P_3	0.9129	0.9159	0.9105	0.9123	23,730,117	8.06 ms
Spatial Attention	P_3	0.9419	0.9453	0.9396	0.9419	23,598,056	8.17 ms
Channel+Spatial	P_2	0.9328	0.9349	0.9315	0.9326	23,631,368	8.03 ms
Channel+Spatial	P_3	0.9437	0.9469	0.9389	0.9420	23,730,216	8.37 ms
Channel+Spatial	P_4	0.9365	0.9361	0.9378	0.9351	24,124,520	7.87 ms
Self-Attention	P_3	0.9419	0.9432	0.9400	0.9412	24,908,678	7.62 ms
Additional Attention-1 (ECA)	P_3	0.9238	0.9217	0.9246	0.9222	23,597,962	7.77 ms
Additional Attention-2 (CBAM)	P_3	0.9201	0.9206	0.9203	0.9194	23,730,216	7.59 ms

Observations: Complexity Analysis

Table 10: Complexity of all 17 models. GFLOPs from the TensorFlow profiler (one image, multiply-add = 2 FLOPs); training time is the measured median step time scaled to the full 2-phase protocol.

Model	Total params	Attn params	Param memory (MB)	GFLOPs / image	Est. training time (min)	Inference (ms/img)
A0 Baseline	23,597,957	0	90	7.75	3.6	8.08
A1 Channel (SE)	23,730,117	132,160	91	7.75	3.7	8.06
A2 Spatial	23,598,056	99	90	7.75	3.7	8.17
A3 Channel+Spatial	23,730,216	132,259	91	7.75	3.8	8.37
A4 Self-Attention	24,908,678	1,310,721	95	8.35	4.1	7.62
A5 Combined (C+S+SA)	25,040,937	1,442,980	96	8.36	4.4	7.93
B1 C+S @ P_1	23,606,520	8,563	90	7.76	5.3	8.16
B2 C+S @ P_2	23,631,368	33,411	90	7.75	4.5	8.03
B4 C+S @ P_4	24,124,520	526,563	92	7.75	3.5	7.87
C1 C+S @ P_3+P_4	24,256,779	658,822	93	7.75	3.9	7.81
C2 C+S @ P_2+P_3+P_4	24,290,190	692,233	93	7.76	4.7	8.41
D1 Spatial→Channel	23,730,216	132,259	91	7.75	3.8	7.97
E1 C+S, k=3	23,730,136	132,179	91	7.75	3.8	8.01
E2 C+S, k=5	23,730,168	132,211	91	7.75	3.8	8.17
F1 ECA	23,597,962	5	90	7.75	3.8	7.77
F2 CBAM	23,730,216	132,259	91	7.75	3.8	7.59
F3 Coordinate Att.	23,698,469	100,512	90	7.76	3.8	8.32

- **Parameters:** ECA adds 5, spatial attention 99, coordinate attention 100,512, Channel/SE, CBAM and Channel→Spatial about 132,000 (+0.56%), and self-attention 1,310,721 (+5.5%). B4 adds 526,563 because its channel MLP acts on 2048 channels.
- **Computation:** only the self-attention models add noticeable work: 8.35 GFLOPs versus 7.75 for the baseline (+0.60, +7.8%). All other modules change the cost by under 0.01 GFLOPs.
- **Training time:** the baseline needs about 3.6 min; most single-module models 3.7–3.8 min; self-attention 4.1 min; the combined model 4.4 min; and attention at P_1 the most (5.3 min), because the modules operate on the largest feature map.
- **Inference time:** all 17 models lie between 7.59 and 8.41 ms/image, and the ordering does *not* follow GFLOPs (e.g. self-attention 7.62 ms, CBAM 7.59 ms, baseline 8.08 ms). Differences of ±0.5 ms are therefore measurement noise on a shared Colab GPU, and no latency conclusion can be drawn from them; GFLOPs is the more reliable indicator of cost.
- **Memory:** parameter memory is 90 MB for the baseline and 95–96 MB for the self-attention models. Peak-GPU-memory readings were also taken, but the training-memory values increased steadily in the order in which the models were benchmarked (from 1,130 MB to 5,128 MB), which shows that memory held over from earlier models contaminated them. They are therefore **not used** here; a clean measurement would require one fresh process per model.

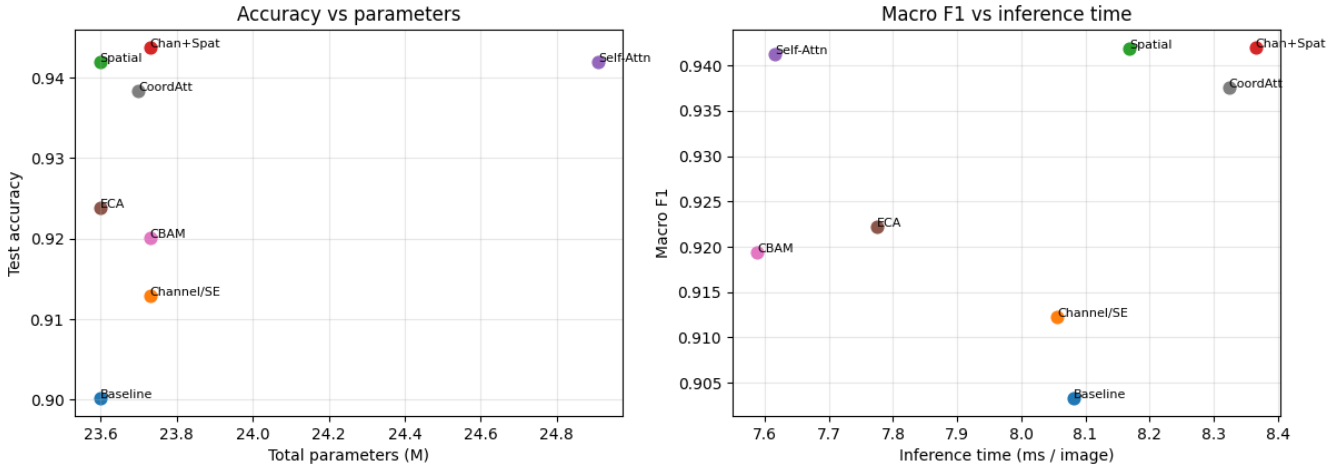


Figure 6: Accuracy versus total parameters (left) and macro F1 versus inference time (right).

Accuracy–complexity trade-off. The non-dominated models (fewest parameters for their accuracy) are the baseline, Spatial (A2), Channel→Spatial (A3) and ECA. **F1–time trade-off:** Spatial, Channel→Spatial, Self-attention and CBAM are non-dominated, but given the timing noise this set should not be over-interpreted. A rank correlation of +0.39 between attention parameters and accuracy over the nine P_3 models shows at best a weak link: the largest module (combined, 1.44 M) is not the most accurate, whereas the 99-parameter spatial module is among the best.

24. Graphical Performance Analysis

Students should not present only numerical tables.

Appropriate graphs must be generated to analyze model behavior.

At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.

Students should compare the convergence behavior of the baseline and attention models.

Observations: Training and Validation Graphs

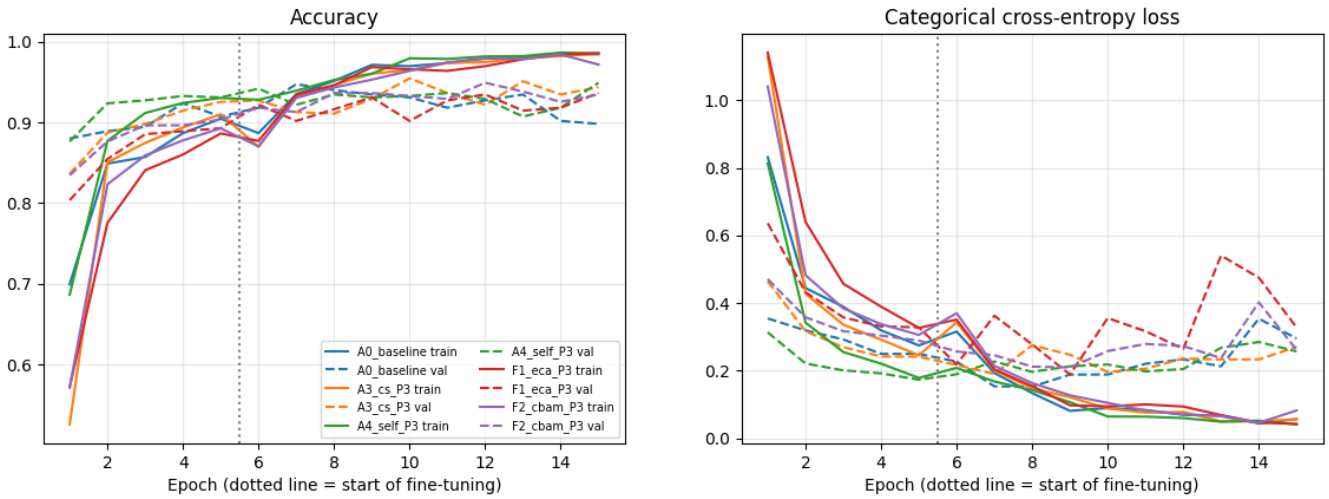


Figure 7: Training (solid) and validation (dashed) accuracy and cross-entropy loss; dotted line = start of fine-tuning.

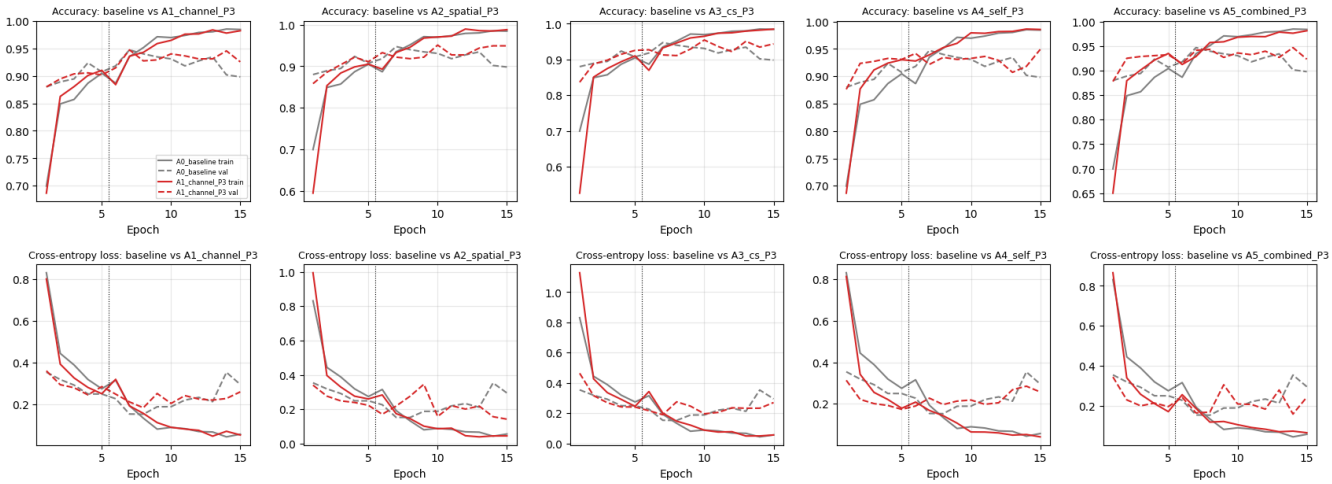


Figure 8: Baseline (grey) versus each attention-type model (red): accuracy (top) and loss (bottom).

Table 11: Observation sheet (Section 38): last-epoch training/validation metrics (epoch 15) and test accuracy for all 17 runs. Training metrics are measured with augmentation and dropout active.

Model	Train acc	Val acc	Test acc	Train loss	Val loss	Train−Val acc	Best-val epoch	Trainable (final)
A0 Baseline	0.9844	0.8982	0.9002	0.0567	0.2952	0.0862	7	14,963,717
A1 Channel (SE)	0.9821	0.9255	0.9129	0.0530	0.2599	0.0566	7	15,095,877
A2 Spatial	0.9879	0.9491	0.9419	0.0456	0.1429	0.0388	10	14,963,816
A3 Channel+Spatial	0.9848	0.9436	0.9437	0.0565	0.2705	0.0412	10	15,095,976
A4 Self-Attention	0.9856	0.9491	0.9419	0.0408	0.2570	0.0365	15	16,274,438
A5 Combined (C+S+SA)	0.9817	0.9236	0.9274	0.0646	0.2431	0.0581	14	16,406,697
B1 C+S @ P_1	0.9910	0.8927	0.9129	0.0346	0.3159	0.0983	14	14,972,280
B2 C+S @ P_2	0.9708	0.9255	0.9328	0.0814	0.1996	0.0454	7	14,997,128
B4 C+S @ P_4	0.9825	0.9309	0.9365	0.0526	0.2445	0.0516	14	15,490,280
C1 C+S @ P_3+P_4	0.9868	0.9382	0.9292	0.0364	0.1955	0.0486	15	15,622,539
C2 C+S @ P_2+P_3+P_4	0.9879	0.9055	0.9074	0.0409	0.3025	0.0825	14	15,655,950
D1 Spatial→Channel	0.9661	0.9364	0.9328	0.1015	0.2357	0.0298	14	15,095,976
E1 C+S, k=3	0.9700	0.9236	0.9056	0.0771	0.2316	0.0464	13	15,095,896
E2 C+S, k=5	0.9755	0.9400	0.9310	0.0846	0.3243	0.0355	10	15,095,928
F1 ECA	0.9856	0.9364	0.9238	0.0429	0.3277	0.0492	15	14,963,722
F2 CBAM	0.9716	0.9345	0.9201	0.0824	0.2613	0.0370	12	15,095,976
F3 Coordinate Att.	0.9782	0.9509	0.9383	0.0695	0.2610	0.0273	14	15,064,165

Observations (Section 24.1/24.2):

- **Convergence speed:** no consistent acceleration. The first epoch with validation accuracy ≥ 0.88 is epoch 2 for

the baseline and for A1–A5; only C1 reaches it in epoch 1, while B1 needs 6 epochs, E1 4 and ECA, CBAM, Coordinate and E2 3.

- **Train–validation gap:** the baseline's gap (0.0862) is larger than that of every attention model except B1 (0.0983); the others range from 0.027 to 0.083, e.g. A2 0.039, A3 0.041, A4 0.037, Coordinate 0.027. Part of this is because the baseline's validation accuracy decays late in training (best at epoch 7, 0.8982 at epoch 15).
- **Overfitting:** the baseline's validation loss rises from its minimum (epoch 8) by 0.14; A2's validation loss is still decreasing at epoch 15 (rise 0.000); by train–validation gap, B1 (0.0983) and the three-stage model C2 (0.0825) overfit most among the attention models, and ECA and E2 show validation-loss rises (0.139, 0.133) similar to the baseline's.
- **Smoothness / instability:** no consistent effect. Epoch-to-epoch validation-accuracy roughness is 0.0130 for the baseline, lower for A1, CBAM and D1 and higher for most others (e.g. A3 0.0170, ECA 0.0194, k=3 0.0368); B4, A4, A5, C1 and E2 have more validation-loss increases (6–7) than the baseline (5). All curves jump when fine-tuning starts (epoch 6).

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM}.$$

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

Observations: Confusion Matrix Analysis

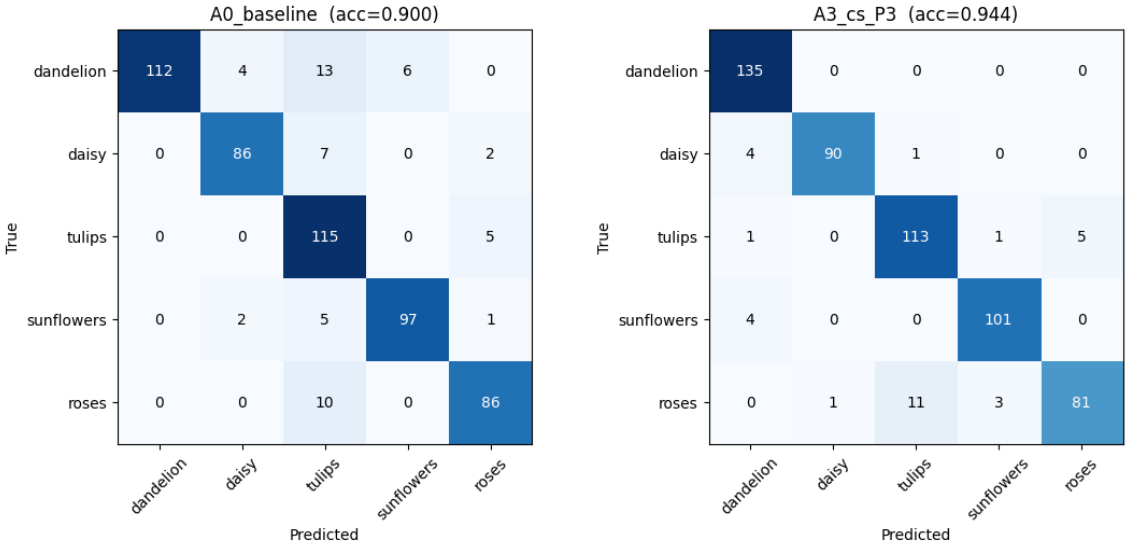


Figure 9: Confusion matrices (551 test images): baseline A0 and selected model A3.

Table 12: Per-class comparison of the baseline (A0) and the selected model (A3, Channel→Spatial at P_3).

Class	Test images	Recall A0	Recall A3	Δ recall	F1 A0	F1 A3	Δ F1
dandelion	135	0.830	1.000	+0.170	0.907	0.968	+0.061
daisy	95	0.905	0.947	+0.042	0.920	0.968	+0.048
tulips	120	0.958	0.942	−0.017	0.852	0.922	+0.071
sunflowers	105	0.924	0.962	+0.038	0.933	0.962	+0.029
roses	96	0.896	0.844	−0.052	0.905	0.890	−0.015

- **Most accurately classified:** tulips for the baseline (recall 0.958), dandelion for A3 (all 135 correct).

- **Frequent confusions:** the baseline mainly predicts *tulips* for other classes (dandelion→tulips 13, roses→tulips 10, daisy→tulips 7) and dandelion→sunflowers 6.
- **What attention reduced:** dandelion→tulips fell from 13 to 0, daisy→tulips from 7 to 1 and dandelion→sunflowers from 6 to 0. Total errors fell from 55 to 31.
- **What it did not:** roses→tulips stayed (10 vs 11) and roses recall fell from 0.896 to 0.844. New, small confusions appeared (sunflowers→dandelion 4, daisy→dandelion 4).
- **Which class benefits most:** tulips (F1 +0.071 for A3) and dandelion (recall +0.170); averaged over all 16 attention variants tulips gains most, whereas roses gains least or loses (Fig. 11).
- **Distribution:** four of five classes improve in F1 with A3, so the gain is broad, but roses does not benefit.

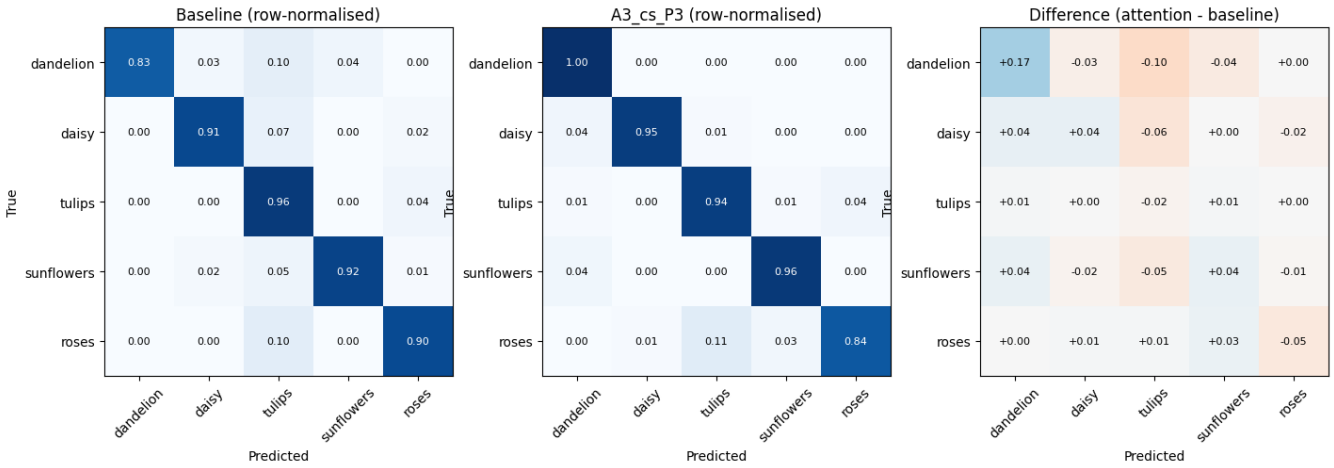


Figure 10: Row-normalised confusion matrices and their difference (attention − baseline).

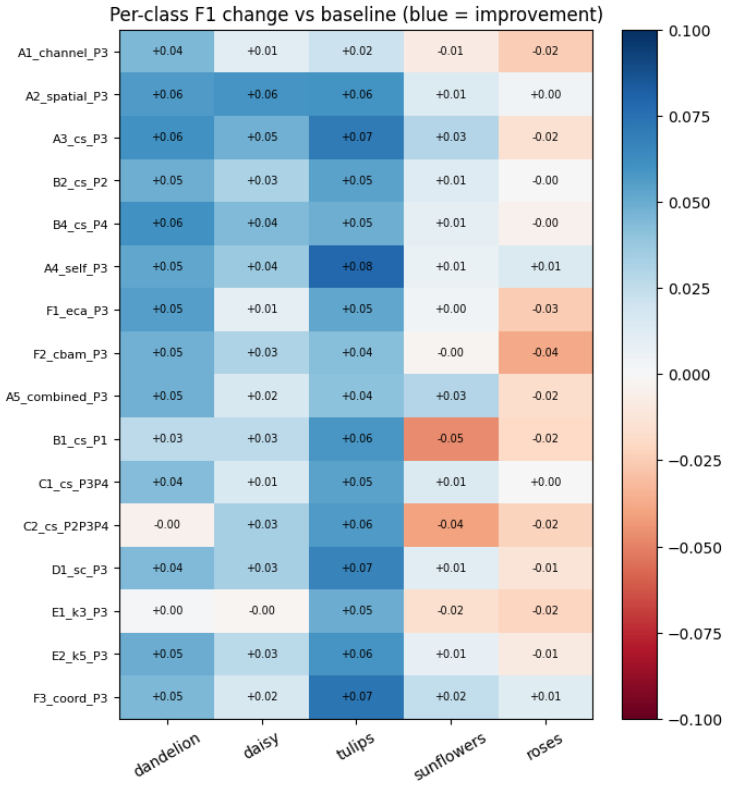


Figure 11: Per-class F1 change relative to the baseline for all attention variants.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

Observations: Test Images and Feature Maps

Five fixed test images were used for all qualitative figures: three correctly classified (tulips, dandelion, roses) and two roses images that the selected model A3 misclassified.

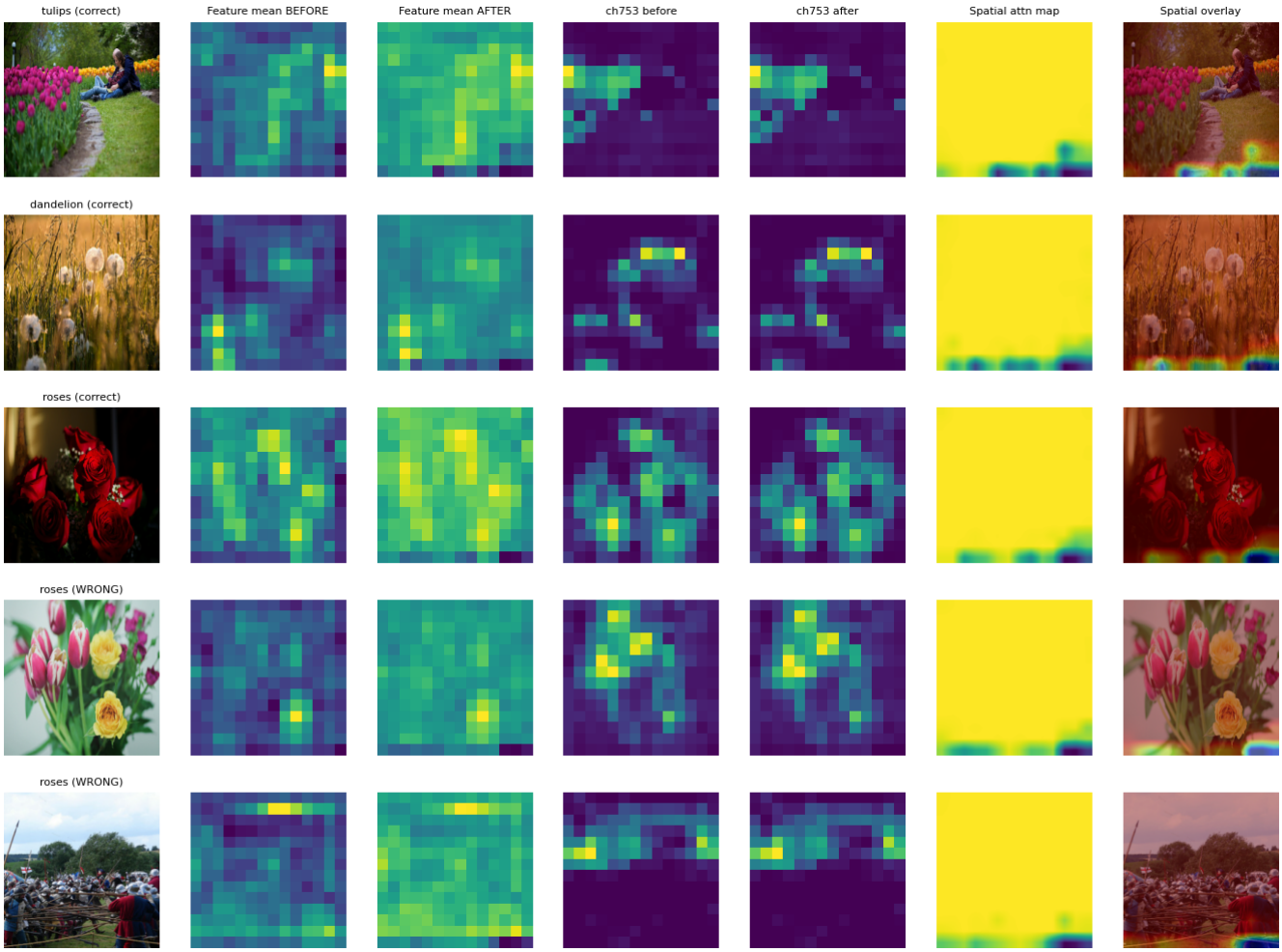


Figure 12: Feature maps before/after the P_3 Channel→Spatial module, selected channel before/after, spatial attention map and overlay for the five test images.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right).$$

The map should be resized to the original image resolution and overlaid on the input.
Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

Observations: Grad-CAM Analysis

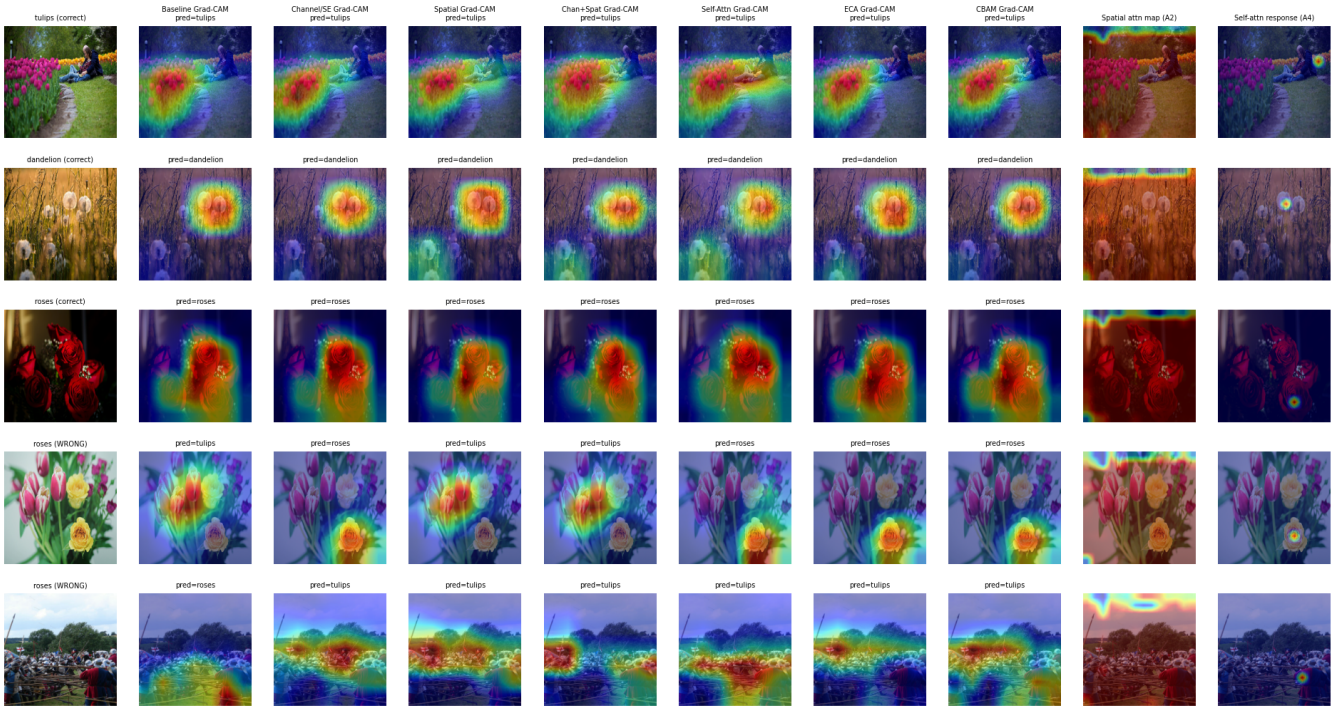


Figure 13: Grad-CAM of baseline, Channel/SE, Spatial, Channel+Spatial, Self-attention, ECA and CBAM models, plus the spatial map of A2 and the self-attention response of A4, for the same five images.

Grad-CAM (post-hoc, computed from gradients of the class score at the last convolutional stage) differs fundamentally from the internal attention maps of Sections 28 and 29, which are computed inside the forward pass. Visible evidence for the five images:

- **Tulips (correct):** all models focus on the dense pink tulip bed on the left; the Channel→Spatial and self-attention models spread somewhat more activation toward the path and the seated people.
- **Dandelion (correct):** every model centres on the two seed heads; spatial, Channel→Spatial and self-attention add a weaker secondary blob on the lower-left dandelion.
- **Roses (correct):** all models concentrate on the central rose cluster; the maps differ only slightly in compactness.
- **Roses, wrong for A3 (bouquet of pink tulips and yellow roses):** the baseline, spatial and Channel→Spatial models predict *tulips* with activation on the pink tulips, whereas the channel, ECA, CBAM and self-attention models predict *roses* with activation on the lower yellow rose. The prediction follows the region the model attends to.
- **Roses, wrong for A3 (crowd scene without a prominent flower):** the baseline predicts roses with activation on the red clothing at the lower right, whereas all attention models predict tulips with a horizontal band of activation across the crowd/ground, i.e. evidence is scene-level and not on a flower; this image is atypical for the class.

Observations from five images are illustrative only; Section 30 checks them on 100 images.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

$$\boxed{\text{Internal Attention}}$$

and

$$\boxed{\text{Post-hoc Explanation}}$$

because they are not the same.

An attention module directly computes weights during the forward pass.

Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.

Instead of displaying every channel when C is large, the student may show the top-K channels.

Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

Observations: Channel Attention

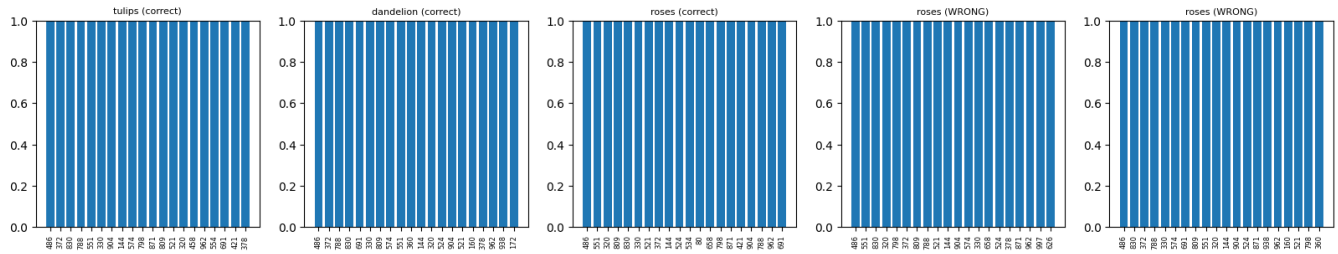


Figure 14: Top-20 channel-attention weights (channel gate of A3 at P_3) for the five test images.

For the channel gate of the selected model A3 (Channel→Spatial at P_3) the weights are strongly bimodal (mean 0.583, standard deviation 0.417 over the 1024 channels); the top-20 channels of every image are all saturated at ≈ 1.0, and the weight vectors of different images are almost identical (mean pairwise correlation **0.997**). Thus a large group of channels receives strong weights, many others are suppressed, and the channel weighting hardly changes with the input: in practice it acts as a learned, nearly static channel selection rather than an image-adaptive one. (For the same reason the "top-K" ranking is unstable: the overlap of the top-20 channels between the two attention orders is only 3–4 of 20, but many channels tie at ≈ 1, so this overlap is not informative.) The feature maps before and after attention look alike for single channels (e.g. channel 753) while the mean feature map after attention is smoother, with more activation on background regions.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

Observations: Spatial Attention

In Fig. 12 and Fig. 16 the spatial maps are ≈ 1 (full pass-through) over most of the image; the low values appear **mainly along the bottom border and in a corner**, not on the flower or the background clutter. This pattern is probably linked to zero padding of the 7×7 convolution at the feature-map border, so the spatial gate acts mostly as a border-suppression (and re-scaling) mechanism rather than as a flower detector. The strong accuracy gain of A2/A3 should be read in this light: it is real in the test numbers, but the maps do not show semantic foreground selection, and the single-run protocol cannot separate the effect of the gate from other effects (for example, regularisation of the fine-tuned conv5_x stage).

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

Observations: Self-Attention

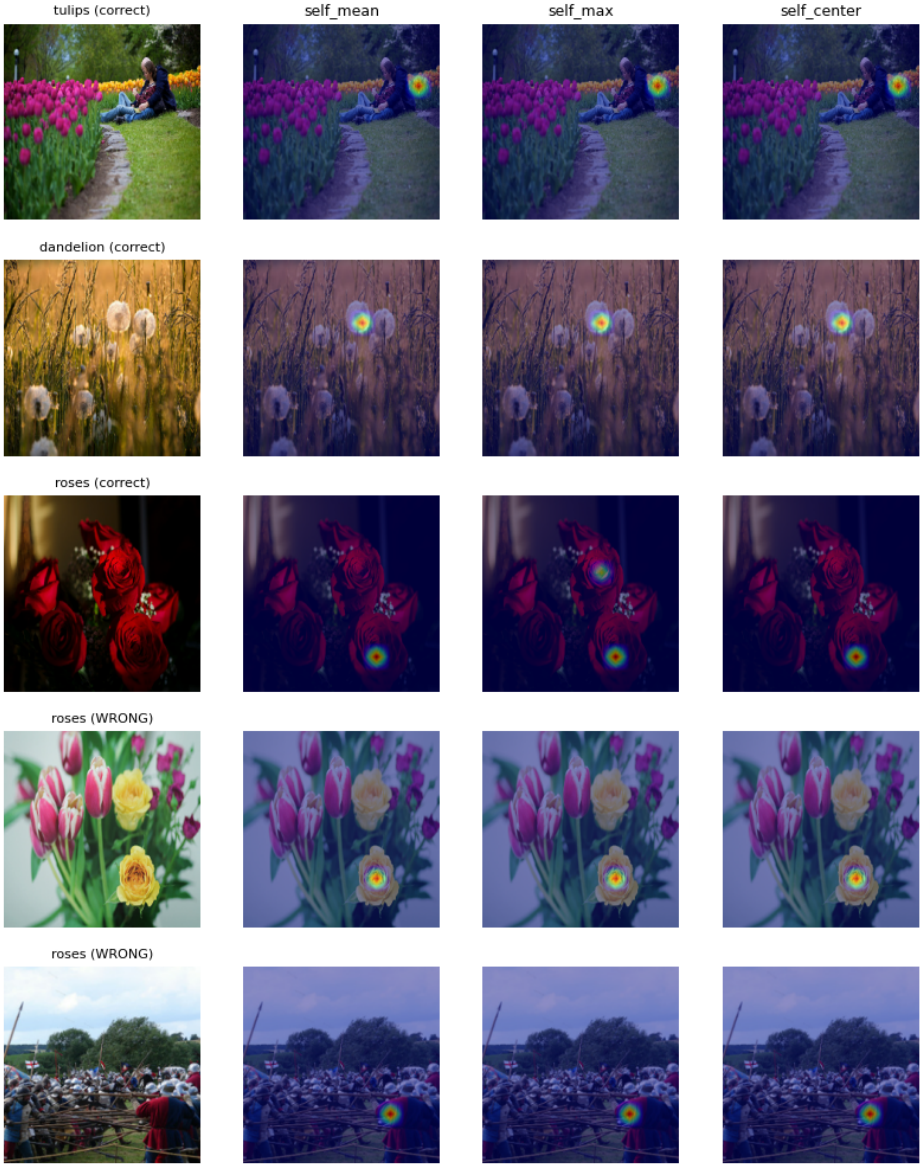


Figure 15: Self-attention responses (mean over queries, maximum over queries, centre query) at P_3.

The learned residual scale reached only $\gamma = 0.059$, so the self-attention branch contributes a small correction. The mean-, maximum- and centre-query responses are almost identical and show a **single compact hot spot** per image (on a dandelion head, on a rose, on the yellow rose in the first wrong image and on the red clothing in the crowd image, but near the person rather than the tulips in the first image). Hence most query positions attend to the same few key positions (a "hub" pattern): distant regions interact through one dominant location rather than through broad pairwise relations. The $N \times N$ matrix has 38,416 entries per image at P_3.

Table 13: Growth of the self-attention matrix with resolution ($d = C/8$, fp32, one head).

Position	Feature map	$N = HW$	$N \times N$ entries	Matrix size / image (MB)	Self-attn params	Self-attn MACs (M)
P_1	$56 \times 56 \times 256$	3136	9,834,496	37.52	81,921	3089.2
P_2	$28 \times 28 \times 512$	784	614,656	2.34	327,681	610.9
P_3	$14 \times 14 \times 1024$	196	38,416	0.15	1,310,721	301.2
P_4	$7 \times 7 \times 2048$	49	2,401	0.01	5,242,881	262.4

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

Observations: Attention at P_2, P_3 and P_4

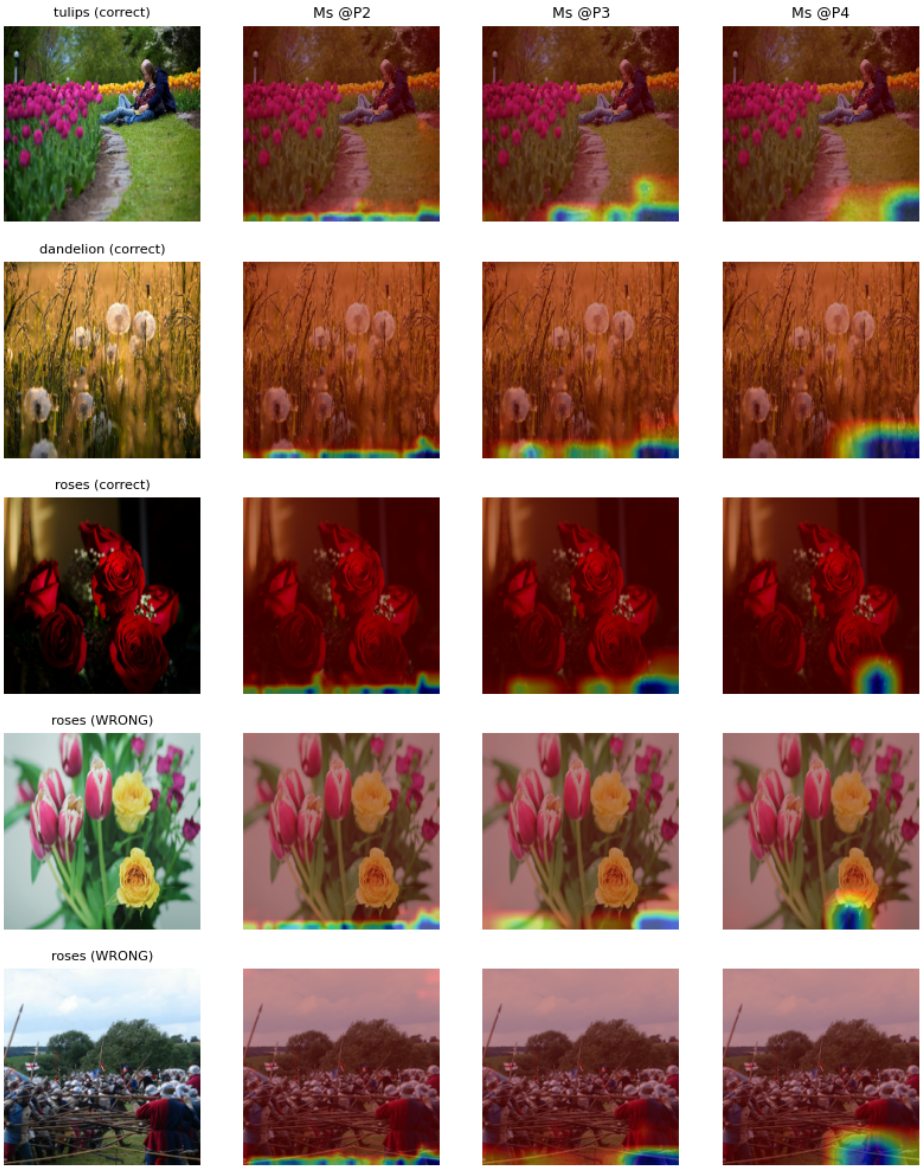


Figure 16: Spatial attention maps $M_s^{P_2}$, $M_s^{P_3}$, $M_s^{P_4}$ for the same five images.

The three maps (from separately trained models) differ mainly in the size and place of the low-valued region: a thin band along the bottom edge at P_2 (28×28), a wider band in the lower right at P_3, and a compact blob in the lower right at P_4 (7×7), consistent with the coarser feature grid. The expectation "P_2 responds to edges and textures, P_4 to larger semantic regions" **is not confirmed** by these figures, because in all three maps the flower regions are

left at ≈ 1. When all stages carry attention in the same network (C2, Fig. 17), the maps change again (vertical low-valued strips at P_3 and P_4), which shows that the maps depend on what the other modules do.

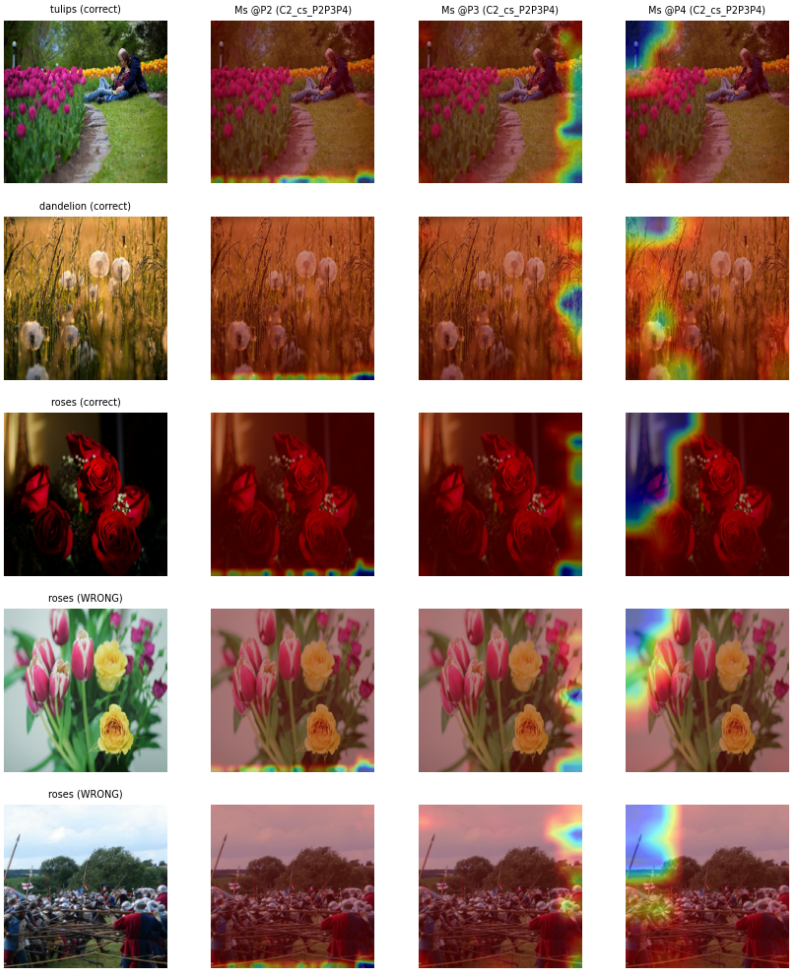


Figure 17: Spatial maps of the three-stage model C2 (P_2+P_3+P_4).

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

Observations: Qualitative Comparison and Localisation Criteria

Table 14: Grad-CAM localisation proxies over 100 fixed random test images (coverage = fraction of pixels $\geq 50\%$ of the CAM maximum; spread = CAM dispersion / image size; centre mass = CAM mass inside the central $50\% \times 50\%$ box, uniform map = 0.25).

Model	Coverage	Spread (lower = compact)	Centre mass	Centre mass (correct)	Centre mass (wrong)	Coverage (correct)	Coverage (wrong)	# wrong
A0 Baseline	0.204	0.308	0.442	0.446	0.382	0.210	0.122	7
A1 Channel (SE)	0.198	0.288	0.475	0.481	0.367	0.200	0.167	5
A2 Spatial	0.195	0.291	0.485	0.489	0.418	0.197	0.158	6
A3 Channel+Spatial	0.212	0.294	0.454	0.458	0.373	0.216	0.154	5
A4 Self-Attention	0.273	0.321	0.420	0.427	0.322	0.281	0.147	6
A5 Combined (C+S+SA)	0.264	0.321	0.428	0.441	0.276	0.265	0.253	8
F1 ECA	0.205	0.284	0.489	0.495	0.399	0.212	0.104	6
F2 CBAM	0.185	0.300	0.445	0.450	0.369	0.189	0.118	6
F3 Coordinate Att.	0.306	0.310	0.440	0.442	0.412	0.315	0.159	6

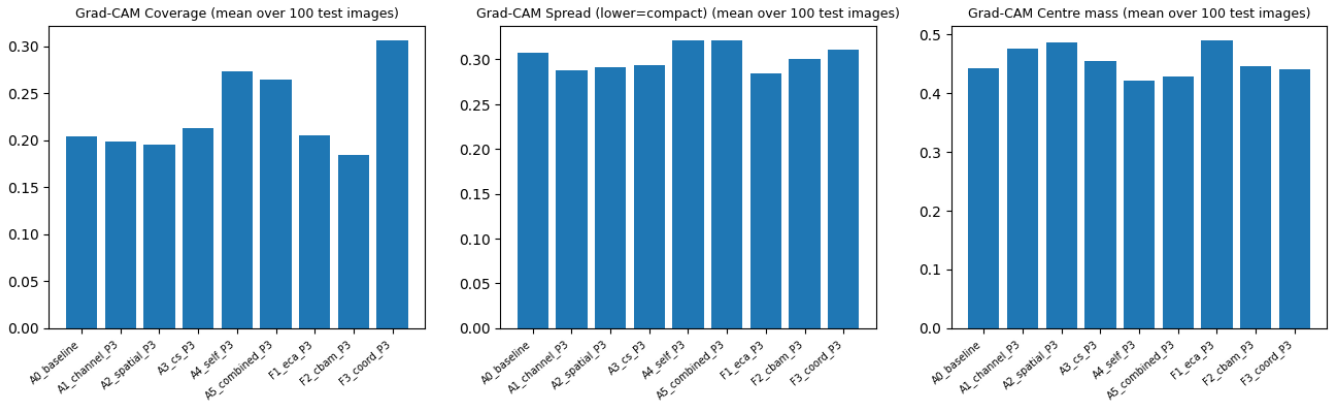


Figure 18: Grad-CAM coverage, spread and centre mass (mean over 100 test images).

Over 100 test images the attention models change where the network looks only slightly: Grad-CAM coverage is 0.204 for the baseline, 0.185–0.212 for channel, spatial, Channel→Spatial, ECA and CBAM, but larger for self-attention (0.273), combined (0.264) and coordinate attention (0.306), i.e. these distribute activation over a wider part of the image. Channel, spatial and ECA models are slightly more compact (spread 0.284–0.291 vs 0.308) and have slightly more mass in the central box (0.475–0.489 vs 0.442); self-attention and combined have less (0.420, 0.428). These shifts are small (3–5 points), so there is **no strong evidence of better foreground focus or background suppression**. For *every* model, Grad-CAM mass in the central box is lower for wrongly classified images (e.g. baseline 0.382 vs 0.446 for correct; A3 0.373 vs 0.458) and coverage is also lower for wrong ones for every model, so incorrect predictions tend to come from less centred, more scattered evidence. Only 5–8 images per model are wrong, so this is indicative. The centre box is a proxy for the foreground, not a segmentation. Differences between shallow (P_2) and deep (P_4) attention are discussed in Section 29; consistency across images is high for the channel weights (0.997) and for the border-type spatial maps.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:
- identical training images;

- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

$$\boxed{\text{SE,} \quad \text{ECA,} \quad \text{CBAM,} \quad \text{Coordinate Attention,} \quad \text{Self-Attention}}$$

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50?
2. What does conv2_x represent compared with conv5_x?
3. Why does the spatial resolution decrease with network depth?
4. Why does the number of feature channels increase with network depth?
5. What is meant by low-level, intermediate, and high-level features?
6. What is the purpose of an attention mechanism?
7. What does channel attention attempt to learn?
8. What does spatial attention attempt to learn?
9. How is channel attention different from spatial attention?
10. Why can channel and spatial attention be used sequentially?
11. Why might channel attention behave differently at conv3_x and conv5_x?
12. Why can spatial attention be useful at an intermediate ResNet stage?
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
15. How does attention performance vary when the insertion position is changed?
16. How does attention at P_2 differ visually from attention at P_4?
17. Does applying attention at multiple stages always improve classification performance?
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
19. How does the spatial-attention kernel size influence the attention map?
20. How does SE attention differ from ECA attention?
21. How does CBAM combine channel and spatial attention?
22. How does self-attention differ from channel attention?
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
24. What is the difference between an internal attention map and Grad-CAM?
25. Why is Grad-CAM called a post-hoc explanation technique?
26. Does the attention model suppress irrelevant background regions?
27. Are the attention maps of correctly classified and incorrectly classified images different?
28. Which flower classes benefit most from attention?
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
30. Why is macro F1 useful in addition to accuracy?
31. What is meant by the accuracy–complexity trade-off?
32. What is meant by the F1–inference-time trade-off?
33. Why must every ablation study use the same dataset split?
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
35. What conclusions can be drawn from the Grad-CAM comparison?

Observations: Answers to the Questions

1. **Residual connections** add the block input to its output ($y = F(x) + x$); the identity path lets gradients reach early layers, avoids degradation in very deep networks and lets blocks learn only a residual correction. ResNet50 has 16 of them.
2. **conv2_x vs conv5_x:** conv2_x ($56 \times 56 \times 256$) holds low-level features (edges, colours, textures) with a small receptive field; conv5_x ($7 \times 7 \times 2048$) holds high-level, class-discriminative semantic features with a large receptive field.
3. **Resolution decreases** because the first block of conv3_x, conv4_x and conv5_x uses stride 2, enlarging the receptive field and reducing computation while building more abstract features.
4. **Channels increase** to encode the richer and more varied patterns needed at higher levels and to compensate for lost spatial detail.
5. **Levels:** low = edges, colour blobs, simple textures; intermediate = repeated patterns, petal arrangements, flower parts; high = whole-object structure and class semantics (daisy vs tulip).
6. **Purpose of attention:** input-dependent re-weighting that emphasises informative channels/locations and suppresses irrelevant ones.

7. **Channel attention** learns which channels matter (one weight per channel from a global descriptor).
8. **Spatial attention** learns which locations matter (one weight per position, shared by all channels).
9. **Difference:** channel attention is spatially uniform ("what"); spatial attention is channel-uniform ("where").
10. **Sequential use** is possible because they are complementary and cheap: the spatial map is computed from already channel-refined features (as in CBAM).
11. **Channel attention at conv3_x vs conv5_x:** conv3_x channels (512) are mid-level pattern detectors, conv5_x channels (2048) are semantic; the module is also 16× larger at conv5_x (526,563 vs 33,411 parameters). In our runs B2 (0.9328) and B4 (0.9365) differ by less than one standard error.
12. **Spatial attention at an intermediate stage** works on a map (28×28 or 14×14) that still resolves flower vs background while the features are already meaningful; P_1 is too low-level (and costly) and P_4 too coarse. In our results P_3 is best and P_1 clearly worse.
13. **Self-attention is expensive at high resolution** because it builds an $N \times N$ matrix with $N = HW$ ($3136^2 \approx 9.8$ M entries per image at P_1).
14. **Matrix size** grows quadratically with N: doubling H and W gives 4× more positions and a 16× larger matrix (9.83 M at P_1 vs 38,416 at P_3).
15. **Position changes performance:** Channel→Spatial accuracy is 0.9129 (P_1), 0.9328 (P_2), 0.9437 (P_3), 0.9365 (P_4).
16. P_2 **vs** P_4 **visually:** a thin bottom-edge band at P_2 versus a compact lower-right blob at P_4; both leave the flower regions at about 1, so the expected edge-vs-semantic distinction is not visible (Section 29).
17. **Multiple stages:** no; P_3+P_4 (−1.5 points) and P_2+P_3+P_4 (−3.6 points) were worse than P_3 alone.
18. **Order:** Channel→Spatial 0.9437 vs Spatial→Channel 0.9328, a difference inside the noise level; maps correlate 0.93–0.99.
19. **Kernel size:** accuracy rises with k (0.9056, 0.9310, 0.9437), but the maps do not show clearly richer context (3×3 maps are the most varied, 5×5 and 7×7 are nearly saturated).
20. **SE vs ECA:** SE uses GAP → bottleneck MLP (reduction r, $\approx 2C^2/r$ parameters, every channel interacts with all others); ECA drops the reduction and applies a 1-D convolution of adaptive kernel size over neighbouring channels (5 parameters at $C = 1024$).
21. **CBAM** applies a channel gate (average- and max-pooled descriptors through a shared MLP, summed, sigmoid) followed by a spatial gate (channel-wise mean and max maps, 7×7 convolution, sigmoid).
22. **Self-attention vs channel attention:** self-attention relates every spatial position to every other (long-range, content-dependent, $N \times N$) and outputs a weighted sum of values; channel attention only rescales channels from a global descriptor.
23. **Largest module = highest accuracy?** No: self-attention (1.31 M) and the combined module (1.44 M) are not better than the 132 k-parameter Channel→Spatial module (0.9437), and the combined module is lower (0.9274).
24. **Internal map vs Grad-CAM:** the internal map is computed by the module in the forward pass and changes the prediction; Grad-CAM is computed afterwards from gradients of a class score and does not change the model.
25. **Post-hoc:** Grad-CAM needs a finished prediction and a backward pass on a trained model; it explains the network but is not part of its computation.
26. **Background suppression:** the spatial gates suppress mainly the image border (bottom band/corner) rather than the background clutter, and Grad-CAM differences are small (Section 30): at most weak evidence.
27. **Correct vs incorrect:** the internal spatial maps look alike for both; Grad-CAM centre-box mass and coverage are lower for incorrect predictions in all nine models (e.g. baseline 0.382 vs 0.446).
28. **Classes that benefit most:** tulips (F1 +0.071 with A3 and the largest average gain) and dandelion (recall 0.830 → 1.000); roses does not benefit.
29. **Confusion matrix vs accuracy:** it shows which classes are confused in which direction (e.g. the baseline's sink class is tulips; attention removed dandelion→tulips but created dandelion-directed errors), per-class recall/precision, and where gains occur, all hidden by one accuracy value.
30. **Macro F1** averages the per-class F1 with equal weight, so weak classes (roses, tulips) are not masked by large ones and precision and recall are balanced; the classes are slightly imbalanced (135 vs 95 test images).
31. **Accuracy–complexity trade-off:** accuracy gained per additional parameter/FLOP; here the 99-parameter spatial module gives nearly the whole gain, so extra parameters bought nothing.
32. **F1–inference-time trade-off:** F1 gained per extra millisecond per image; here latency differences (±0.5 ms) are within measurement noise, so GFLOPs (+0.60 only for self-attention) is the better cost measure.
33. **Same split** so differences come from the model and not from a different test set (one accuracy has a 1.28-point standard error).
34. **Same protocol** (optimiser, learning rates, epochs, augmentation, seed, fine-tuning) because these influence accuracy as much as the module; only the factor under study may change.
35. **Grad-CAM conclusions:** for correct images baseline and attention models look at the same flower; differences appear for wrongly classified images, where the predicted class follows the region highlighted; over 100 images the changes are small (coverage and centre mass shift by 0.02–0.1), so Grad-CAM supports "similar focus, somewhat different coverage" but not "clearly better localisation".

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

Where the Observations Are Recorded

Observation 1 (feature dimensions) is recorded in Section 13; observations 2–6 and 10–12 are in the observation sheet in Section 24 and the complexity table in Section 23; macro precision, recall and F1 are in the tables of Sections 14–19 and 23; confusion matrices are in Section 25; channel-attention weights in Section 28.1; spatial maps in Sections 28.2 and 29; self-attention responses in Section 28.3; Grad-CAM maps in Section 27; and attention maps for different insertion positions in Section 29.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;

- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

Observations: Discussion

Why did attention change performance? Three of the six attention-type models (spatial, Channel→Spatial, self-attention) improved the baseline by about four points and the channel-only modules by 1–2 points. Because the channel weights are almost input-independent and the spatial maps pass most of the image unchanged, the improvement is unlikely to come from careful selection of flower regions; plausible contributors are re-scaling of the features entering conv5_x (which is fine-tuned), reduced late overfitting (the baseline's validation accuracy fell after epoch 7) and run-to-run variation. A multi-seed study would be needed to separate these. **Position and resolution.** The best position (P_3) has moderate resolution and meaningful features; P_1 is both less accurate and most expensive. **Complexity.** Only self-attention increases GFLOPs noticeably and its cost scales with $(HW)^2$, which is why it belongs at P_3/P_4. **Do improvements justify cost?** For the spatial and Channel→Spatial modules yes (negligible cost); for the combined and multi-stage models no. **Do quantitative and qualitative results agree?** Partly: both show that attention changes behaviour only modestly (small Grad-CAM shifts), while accuracy changes by up to 4 points; the internal maps reveal border-type spatial gating and nearly static channel gating, which explains why strong "semantic" attention was not observed.

Observations: Limitations

(i) Each configuration was trained once (seed 42); differences below about 2.5 points are not reliable. (ii) Last-epoch weights were used without checkpoint selection. (iii) Inference times are noisy; peak-memory readings were contaminated and are not reported. (iv) The centre-box measure is a proxy for the foreground. (v) Qualitative statements rest on five images (plus 100-image proxies); one of the "wrong" images contains no prominent flower.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

Observations: Conclusion

Under an identical protocol, attention inserted after conv4_x of a pretrained ResNet50 raised test accuracy on `tf_flowers` from 0.9002 to as much as 0.9437 (Channel→Spatial; spatial-only and self-attention 0.9419), with macro F1 from 0.9033 to 0.9420. The best position is P_3; attention at P_1 helps little, and attention at several stages, a combined C+S+SA module or a larger parameter budget did not help. The order of channel and spatial attention has no reliable effect, and larger spatial kernels gave higher accuracy without clearly richer maps. Computationally only self-attention is noticeably heavier (+0.60 GFLOPs, +1.31 M parameters), while spatial attention with 99 parameters performs equally well. Internal maps show nearly static channel weights and border-dominated spatial gates, and Grad-CAM shows similar focus for baseline and attention models with small changes in coverage, so the gains cannot be attributed to markedly better flower localisation. The overall conclusion rests on a single-seed study and should be confirmed with repeated runs.

42. References

1. K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
2. J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-Excitation Networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 8, pp. 2011–2023, 2020.
3. S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module," *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 3–19, 2018.
4. Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11534–11542, 2020.
5. Q. Hou, D. Zhou, and J. Feng, "Coordinate Attention for Efficient Mobile Network Design," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13713–13722, 2021.
6. A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
7. R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.
8. TensorFlow Documentation: https://www.tensorflow.org/
9. Keras Documentation: https://keras.io/
10. TensorFlow Datasets: https://www.tensorflow.org/datasets